Indranil Paul

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WORK EXPERIENCE

Software Engineer (Python, PyTorch, Scikit-learn, RAG, LLMs)
Airawat Research Foundation, Kanpur Jul 2025 – Mar 2026

- Designed **end-to-end ML pipelines** for **AQI predictive analytics** using **time-series** IoT sensor data from 100+ stations across UP and Bihar
- Applied **Transformer-based architectures** and **hierarchical clustering** for pollution source profiling with **anomaly detection** on high-volume sensor streams
- Finetuned **LLM via supervised fine-tuning and preference alignment (RLHF)** and scalable **feature engineering workflows** for **real-time analytics** and reporting
- Performed **statistical modeling** and **exploratory data analysis (EDA)** to uncover airshed boundary patterns and validate sensor data integrity

Data Engineer / Analyst (Informatica, Tableau, PostgreSQL)
EMC², Bengaluru *Feb 2017 – Nov 2022*

- Built **30+ Informatica ETL/ELT pipelines** with Python automation, processing **10M+ records/day** with robust **data quality validation**
- Developed **10+ Tableau BI dashboards**, reducing reporting effort by **40%** and data latency by **35%** via **real-time data pipelines**

EDUCATION

MTech in Computer Science, *IIT Kanpur* 2023–2025
Supervisor: **Prof. Piyush Rai** — CGPA: **8.57/10** — Coursework: **Probabilistic Machine Learning, Deep Generative Models, Probability Theory, Deep Reinforcement Learning, Mathematics for CS**

BTech in Computer Science, *Government College of Engineering and Leather Technology* 2012–2016
CGPA: **8.18/10**

TECHNICAL SKILLS

- **Programming:** **Python**, **C/C++**, **SQL**, **PostgreSQL**
- **Machine Learning:** **Supervised & Unsupervised Learning**, **Statistical Modeling**, **A/B Testing**, **Model Evaluation & Optimization**, **Feature Engineering**
- **Deep Learning:** **Transformers**, **CNNs**, **RNNs/LSTMs**, **Vision Transformers**, **Transfer Learning**, **Generative Models (VAE, GAN, Diffusion)**
- **Information Retrieval:** **Ranking Systems**, **Recommendation Systems**, **Search/Relevance Modeling**
- **Data & Distributed Systems:** **Large-scale Data Processing**, **ETL Workflows**
- **Lib & Frameworks:** **PyTorch**, **Scikit-learn**, **Hugging Face Transformers**, **NumPy**, **Pandas**, **Manim**
- **LLM Systems:** **RAG Pipelines**, **Fine-tuning**, **LangChain**, **Prompt Optimization**, **PEFT**.
- **Tools:** **Git**, **Jupyter**, **VSCode**, **Databricks**

ACHIEVEMENTS

- **Reviewer: ICLR 2026**(Reviewed research papers on **interpretability** and **Explainable AI**), **ICLR 2027**
- GATE CS: **AIR 156 (2023)**, **AIR 390 (2021)**
- Secured **AIR 7 in ISI Kolkata MTech CS 2022**

PROJECTS

NTK-JL: A Unified Kernel Framework for Training-Data Attribution *Under Review, TMLR* — *OpenReview*
PyTorch, Neural Tangent Kernel, Representer Theorem, Kernel Ridge Regression

- Developed **NTK-JL**, a retrain-free training-data attribution framework that linearises a fine-tuned model via the **empirical Neural Tangent Kernel** and solves kernel ridge regression in a **Johnson–Lindenstrauss-sketched** space, cutting gradient storage from $O(nP)$ to $O(ns)$.
- Proved the unified kernel admits **two readouts** that provably recover two independently-developed attribution methods as special cases: a decomposition readout reducing to **Representer Point Selection** (Spearman ρ=0.97 vs. frozen ViT-B/16) and a datamodel readout **algebraically identical to single-model TRAK**.
- Used the **Linear Datamodeling Score** to show the two readouts answer different questions: the decomposition readout has near-zero counterfactual fidelity, while the datamodel readout matches TRAK-level fidelity (**LDS 0.63–0.75** on SST-2/QNLI).
- Showed via proponent/opponent deletion on **DistilBERT/SST-2** and **RoBERTa-base/QNLI** that NTK-JL gradients localise task-specific signal missed by frozen-feature RPS (**AUC-DEL+ gaps of 14–20 pp**).
- Derived self-influence scores at no additional cost, surfacing probable **annotation errors and labeling ambiguity** at **2.8–3.2× enrichment** over random on QNLI and SST-2.
- Connected the datamodel readout to a **Gaussian-process posterior-sensitivity** interpretation via the primal–dual kernel ridge identity, framing TRAK-style attribution as inverse-covariance propagation, not an ad hoc heuristic.

Semantic Book Recommendation using LLM
Python, Retrieval Systems, Vector Search, Transformers Sep 2025 — GitHub

- Built an **end-to-end semantic retrieval & recommendation system** over **5.2K+ books** using **Transformer embeddings** and **ChromaDB** vector search, achieving **sub-second query time**
- Combined semantic similarity, zero-shot **genre normalization** (**BART-MNLI**, 78% acc.) and **emotion-aware filtering** (**RoBERTa**, 76% acc.) for multi-stage ranking, reducing label space from 500+ categories to **4 canonical genres**
- Built an interactive **Gradio** interface supporting free-text search, faceted filtering, and real-time recommendations

Computer Vision using Transfer Learning and Vision Transformer (ViT)
PyTorch, Deep Learning, Computer Vision May 2024 — GitHub

- Built **transfer learning pipelines** for image classification on **FoodVision Mini** using **EfficientNet-B0/B2**, achieving **83–85% accuracy** with **60% reduced training time**
- Conducted a **systematic CNN vs. Transformer comparison**, showing **ViT-Base** reaches **93%+ accuracy** (**+5–7% F1**) over CNN baselines with superior data efficiency under limited-data regimes
- Optimized training via **fine-tuning strategies** (layer freezing, LR scheduling), improving convergence stability and reducing overfitting